$$\lim_{x\to\infty} \log g(x) = \lim_{x\to\infty}\left(\frac{\log(1+2x)}{2\log x}\right) \quad \left(\frac{\infty}{\infty} \text{ form}\right)$$

$$= \lim_{x\to\infty}\left(\frac{\frac{2}{1+2x}}{\frac{2}{x}}\right) \quad \text{(by l'Hôpital Rule)}$$

$$= \lim_{x\to\infty}\left(\frac{x}{1+2x}\right) \quad \left(\frac{\infty}{\infty} \text{ form}\right)$$

$$= \lim_{x\to\infty}\left(\frac{1}{2}\right) = \frac{1}{2} \quad \text{but,}$$

$$\lim_{x\to\infty} \log g(x) = \log\left(\lim_{x\to\infty} g(x)\right).$$

Hence by exponentiating, we get the required limit as $\sqrt{e}$.

Example 7.45

Evaluate : $\lim_{x\to 1} x^{\frac{1}{1-x}}$.

Solution

Let $g(x) = x^{\frac{1}{1-x}}$. This is an indeterminate of the form 1^{∞} when $x\to 1$. Taking the logarithm,

$$\log g(x) = \frac{\log x}{1-x}.$$

$$\text{Therefore, } \lim_{x\to 1} \log g(x) = \lim_{x\to 1}\left(\frac{\log x}{1-x}\right) \quad \left(\frac{0}{0} \text{ form}\right).$$

An application of l'Hôpital rule, gives

$$\lim_{x\to 1}\left(\frac{\frac{1}{x}}{-1}\right) = -1.$$

$$\text{But, } \lim_{x\to 1} \log g(x) = \log\left(\lim_{x\to 1} g(x)\right).$$

Hence on exponentiating, we get

$$\lim_{x\to 1} x^{\frac{1}{1-x}} = e^{-1} = \frac{1}{e}.$$

EXERCISE 7.5

Evaluate the following limits, if necessary use l'Hôpital Rule :

1. $\lim_{x\to 0} \frac{1-\cos x}{x^2}$
2. $\lim_{x\to\infty} \frac{2x^2-3}{x^2-5x+3}$
3. $\lim_{x\to\infty} \frac{x}{\log x}$
4. $\lim_{x\to \frac{\pi}{2}^-} \frac{\sec x}{\tan x}$
5. $\lim_{x\to\infty} e^{-x}\sqrt{x}$
6. $\lim_{x\to 0}\left(\frac{1}{\sin x} - \frac{1}{x}\right)$
7. $\lim_{x\to 1^+}\left(\frac{2}{x^2-1} - \frac{x}{x-1}\right)$
8. $\lim_{x\to 0^+} x^x$
9. $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x$

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